Abu Dhabi City Campus Daily/Weekly Planner

Week 9: 26th – 30th October 2026

Subject: Mathematics Pre-AP Geometry with Statistics **Unit Title: Inference from Data** **Grade: 9**

Lessons	Learning targets	Assessment tools and strategies Formative / Summative / Differentiation	Resources and links for Parents
Lesson 1: **Sample Size & Representation**	I can explain how sample size affects how well a sample represents its population.	5.3 Use graduated examples before students compare simulated samples.	Pre-AP Geometry, Lesson 1.3.4 Identify biases in sampling methods for experiments, observational st
Lesson 2: **Methods of Gathering Data (Experiment)**	I can identify when an experiment is an appropriate method for gathering data and explain how manipulating an independent variable supports an investigation.	2.3 Use a decision tree and annotated scenarios to represent how an experiment is structured.	
Lesson 3: **Observational Studies & Surveys**	I can distinguish an observational study from a survey and explain when each method is appropriate.	5.3 Provide a model comparison before students analyse unfamiliar scenarios. 2.3 Represent the methods with a comparison table showing researcher action, data source and suitable purpose.	
Lesson 4: **Choosing an Appropriate Data Collection Method**	I can select and justify an experiment, observational study or survey for a given purpose.	8.2 Sequence scenarios from familiar school contexts to less familiar multi-constraint situations while maintaining grade-level reasoning demand.	
Homework	Sample Size & Representation, Methods of Gathering Data, Observational Studies & Surveys and Appropriate Data Collection Method.		
Teacher's notes / Reminders	Students should bring a device, scientific calculator and notebook and have access to the PPTs.		

Technology Integration	Microsoft Teams and OneNote; random-sampling simulations; spreadsheet/GDC; population/sample diagrams; research-design templates; survey examples; scenario-sort cards.
Cross-curricular links	Science: choosing experiments and observational studies; identifying variables and considering fair tests. Social Sciences/Civics: surveys, sampling and understanding community perspectives. Business/Economics: customer surveys and evidence-based decisions. ICT/Data Science: digital sampling simulations, organising survey responses and comparing sample results.
External Assessment Integration	Direct alignment with College Board Pre-AP Geometry with Statistics
Wellbeing/ ATL Skills	**Thinking:** compare samples, analyse representativeness, classify data-gathering methods, evaluate design choices and identify limitations. **Communication:** justify conclusions using statistical vocabulary, evidence and clear explanations. **Self-management:** follow investigation steps, organise information, check assumptions and act on feedback. **Social:** collaborate on scenario analysis and give respectful, evidence-based peer feedback.
Standards	Pre-AP Geometry with Statistics – Unit 1: Measurement in Data. Key Concept 1.3: Inferences from Data. 1.3.2: Describe how sample size impacts how well a sample represents the population from which it was drawn. 1.3.3: Design a method for gathering data appropriate for a given purpose; distinguish experiments, observational studies and surveys.
Inquiry/ TOK Question	How can we decide whether a sample gives a trustworthy picture of a population, and how should the research question guide the method we use to collect data?
Success Criteria (Measurable)	**Lesson 1: Sample Size & Representation** • **All students must** explain the difference between a sample and a population. • **Most students should** use evidence from simulated samples to compare how sample size affects representation. • **Some students could** explain why a larger sample may still be unrepresentative if the selection process is biased. **Lesson 2: Methods of Gathering Data (Experiment)** • **All students must** identify an experiment from a scenario. • **Most students should** identify the manipulated independent variable and explain the purpose of the experiment. • **Some students could** design a simple experiment and identify possible sources of bias or limitations. **Lesson 3: Observational Studies & Surveys**

	• **All students must** distinguish an observational study from a survey. • **Most students should** justify which method fits a given research question. • **Some students could** compare strengths and limitations of both methods for the same question. **Lesson 4: Choosing an Appropriate Data Collection Method** • **All students must** select an experiment, observational study or survey for a given purpose. • **Most students should** justify the choice using the research question, variables and type of information needed. • **Some students could** identify a limitation or potential source of bias and propose an improvement.
National Identity Domain	Belonging / Citizenship: Use school and UAE community questions that invite students to investigate shared experiences responsibly. Respect: Students consider different perspectives when designing surveys and discussing results. Responsible decision-making: Students select data-collection methods that suit the question and avoid making claims beyond what the data can support.
	Area: Responsible decision making. Implementation: Students justify their choice of data-collection method, acknowledge limitations and consider how sampling or method choices may introduce bias.